Data visualization

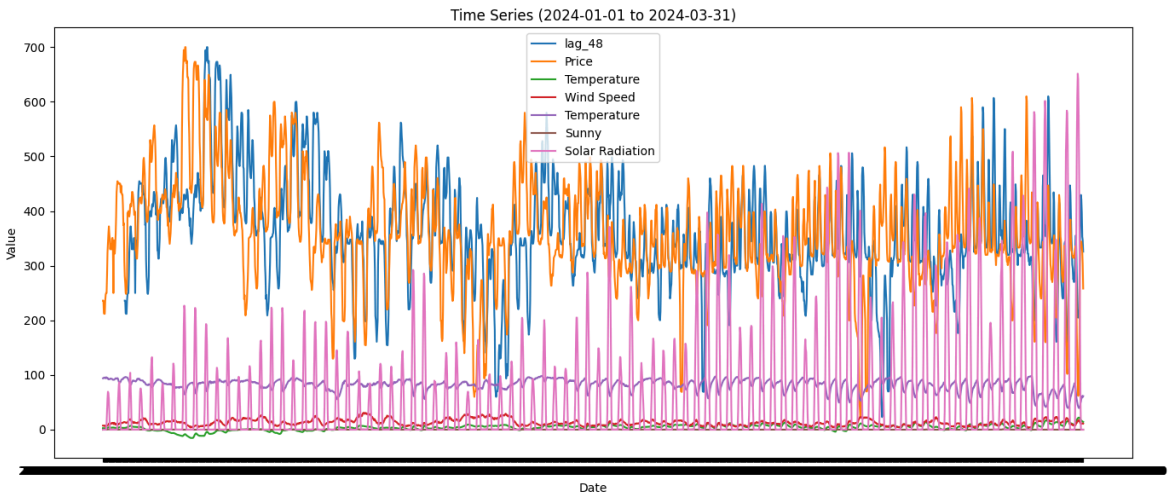
National data

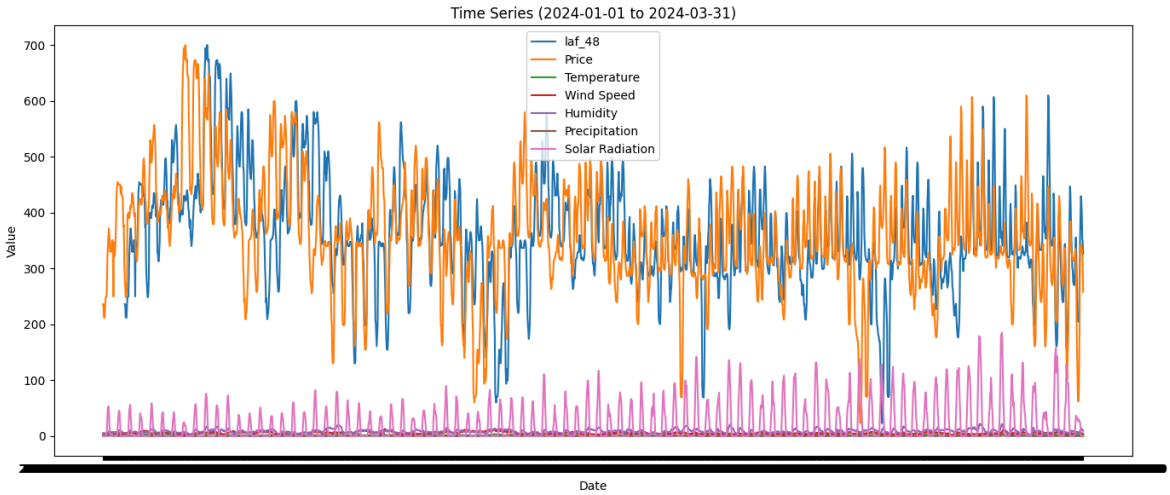
First we can plot the data for the time series on national data, with the relative correlations. In the case of te national features is easier to print all the possible correlations in different forms.

In [49]:
```
import pandas as pd

df = pd.read_csv("../data/output/national_features.csv")
df = df.set_index("delivery_timestamp")
df.rename(columns={"fixing_i_price_first": "fixing_i_price", "business_date_firs

# choose a specific time window
start, end = "2024-01-01", "2024-03-31"
df_slice = df.loc[start:end]
```

Time series plot

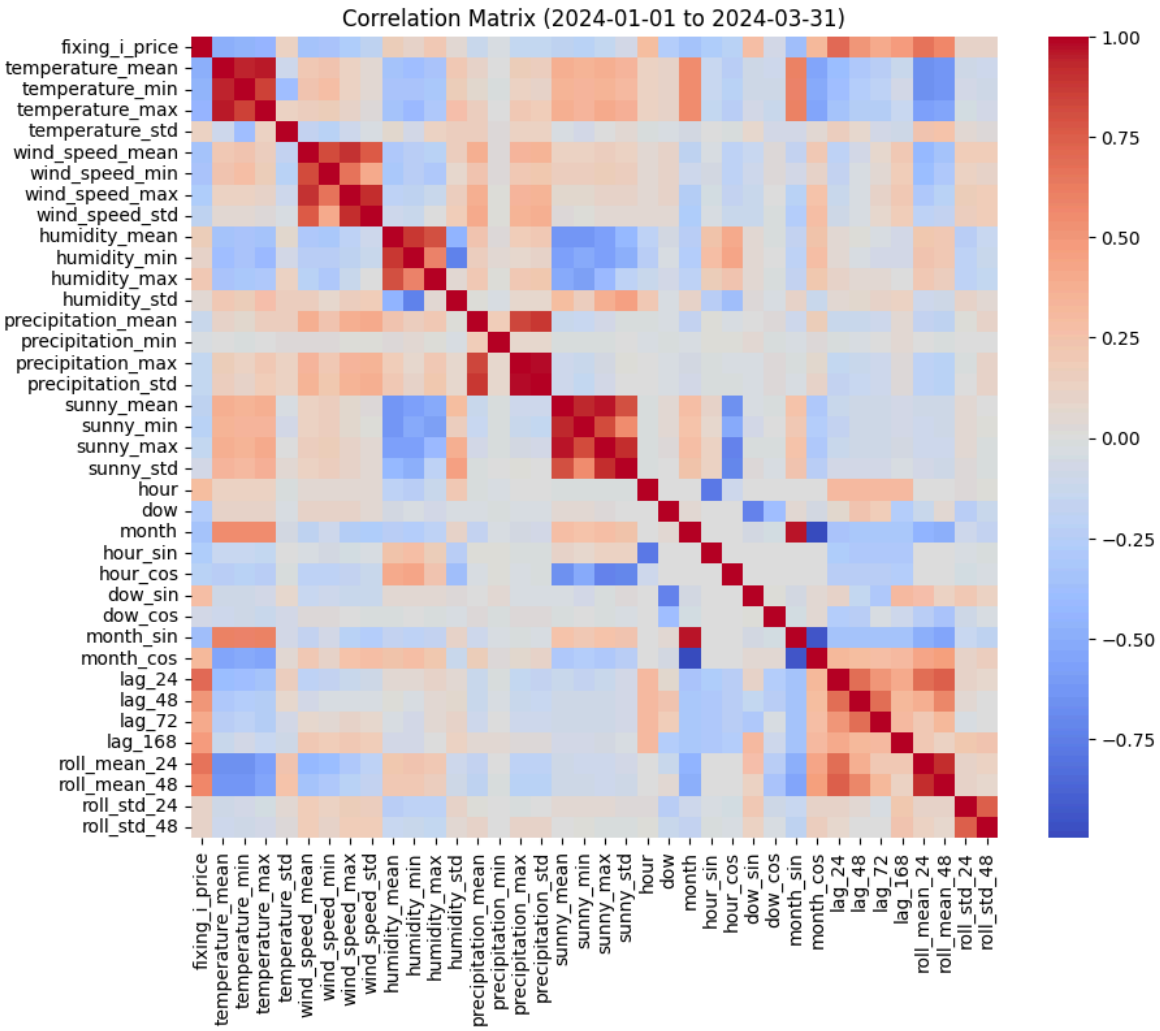
In [52]:
```
import matplotlib.pyplot as plt

plt.figure(figsize=(14, 6))
plt.plot(df_slice.index, df_slice["lag_48"], label="lag_48")
plt.plot(df_slice.index, df_slice["fixing_i_price"], label="Price")
plt.plot(df_slice.index, df_slice["temperature_mean"], label="Temperature")
plt.plot(df_slice.index, df_slice["wind_speed_mean"], label="Wind Speed")
plt.plot(df_slice.index, df_slice["humidity_mean"], label="Temperature")
plt.plot(df_slice.index, df_slice["precipitation_mean"], label="Sunny")
plt.plot(df_slice.index, df_slice["sunny_mean"], label="Solar Radiation")
plt.title("Time Series ({} to {})".format(start, end))
plt.xlabel("Date")
plt.ylabel("Value")
plt.legend()
plt.tight_layout()
plt.show()

plt.figure(figsize=(14, 6))
plt.plot(df_slice.index, df_slice["lag_48"], label="laf_48")
plt.plot(df_slice.index, df_slice["fixing_i_price"], label="Price")
plt.plot(df_slice.index, df_slice["temperature_std"], label="Temperature")
plt.plot(df_slice.index, df_slice["wind_speed_std"], label="Wind Speed")
plt.plot(df_slice.index, df_slice["humidity_std"], label="Humidity")
plt.plot(df_slice.index, df_slice["precipitation_std"], label="Precipitation")
plt.plot(df_slice.index, df_slice["sunny_std"], label="Solar Radiation")
plt.title("Time Series ({} to {})".format(start, end))
plt.xlabel("Date")
plt.ylabel("Value")
plt.legend()
plt.tight_layout()
plt.show()
```





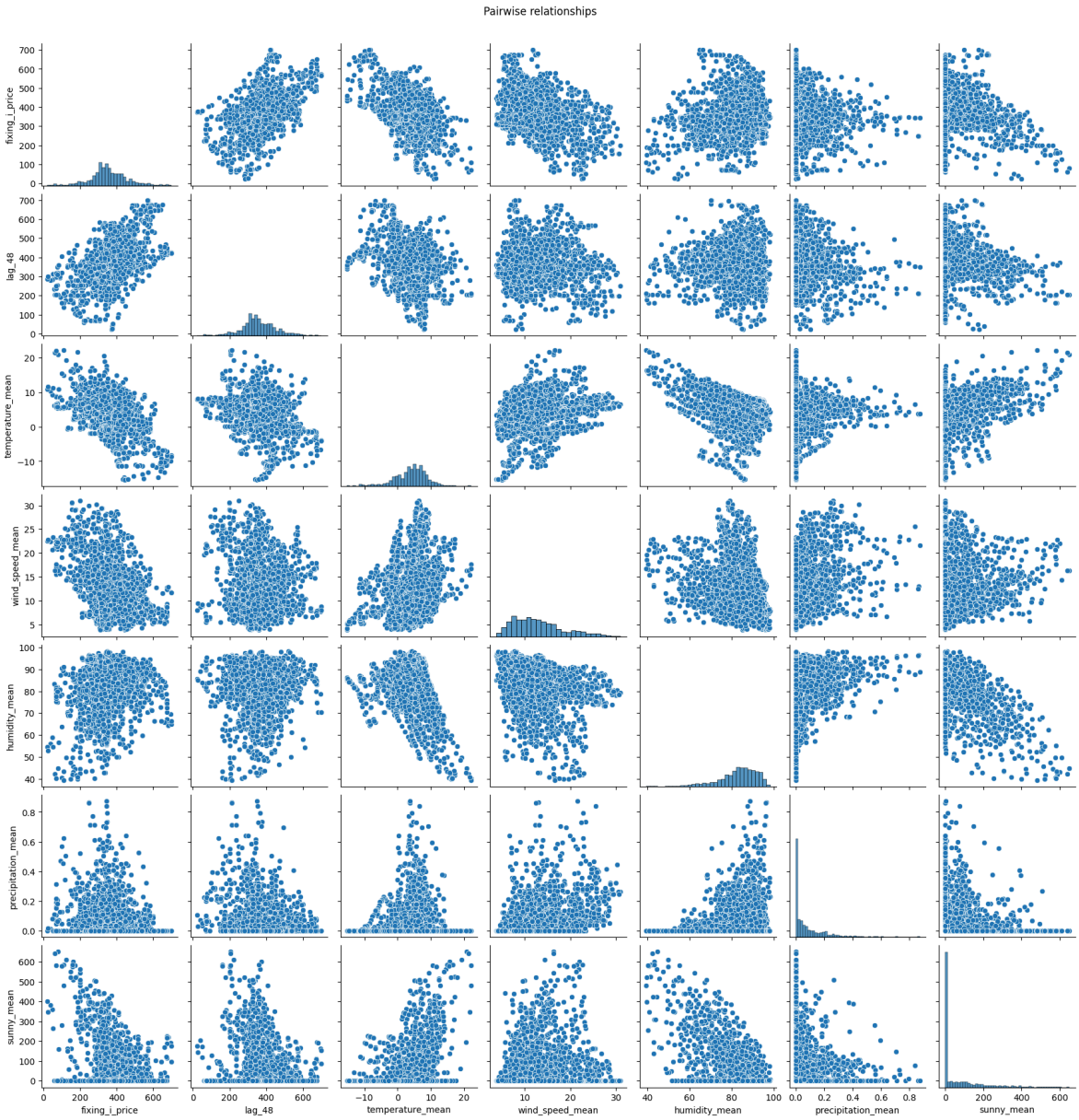
Correlation matrix

In [50]:
```
import seaborn as sns

corr = df_slice.loc[:, (df_slice.columns != "delivery_timestamp") &
                       (df_slice.columns != "business_date")].corr()

plt.figure(figsize=(10, 8))
sns.heatmap(corr, cmap="coolwarm")
plt.title("Correlation Matrix ({} to {})".format(start, end))
plt.show()
```



Pairwise relationships

In [55]:
```
sns.pairplot(df_slice[["fixing_i_price", "lag_48", "temperature_mean",
                       "wind_speed_mean", "humidity_mean", "precipitation_mean",
plt.suptitle("Pairwise relationships", y=1.02)
plt.show()
```



City features

Now we can try to print some plots related to the single cities data.

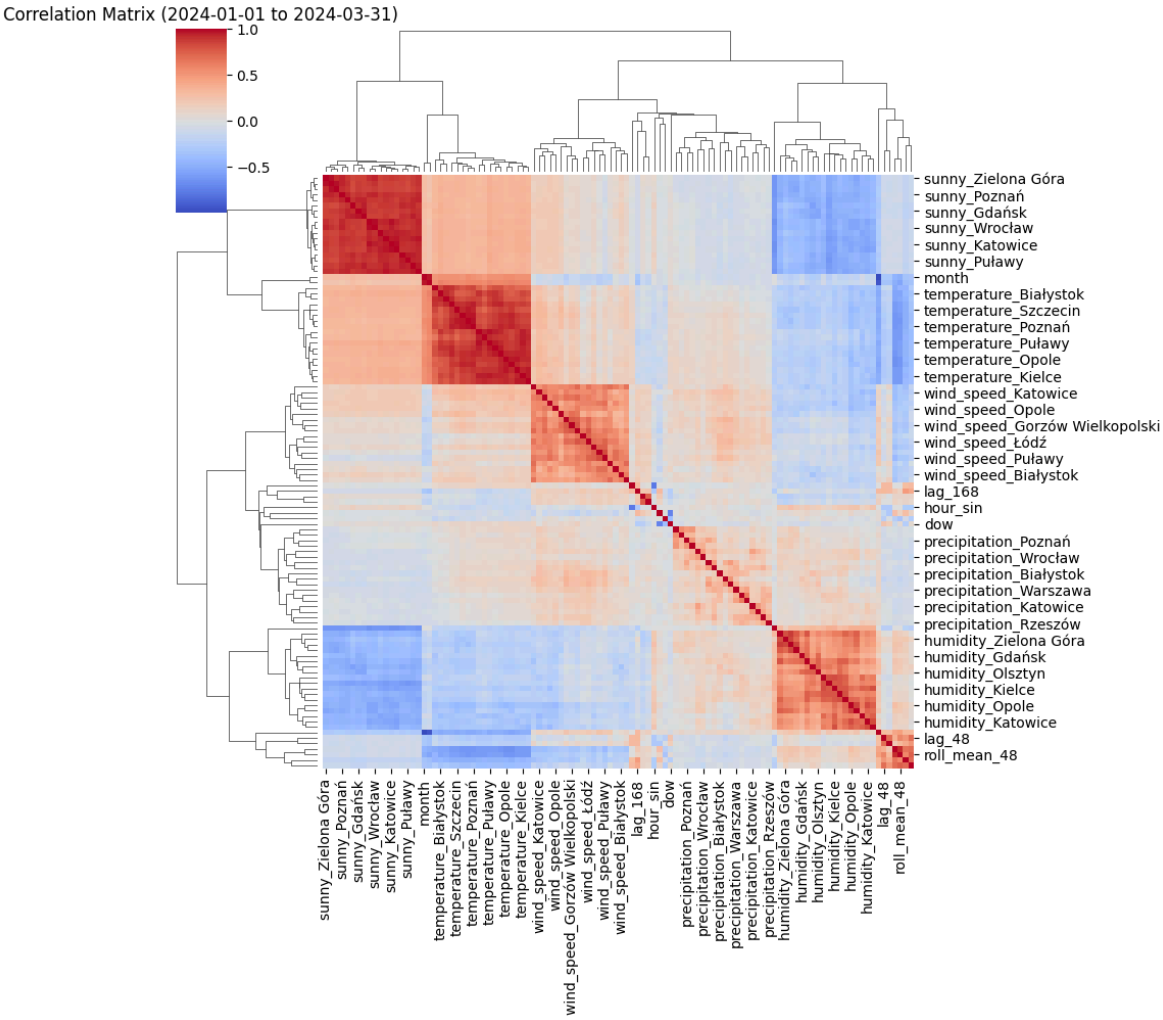
Due to the eleveted number of features it would be pointless to print a time series plot, so we could think to print the correlation matrix, that can give us a general idea of the correlation between cities and prices.

In [56]:
```
import pandas as pd

df = pd.read_csv("../data/output/city_features.csv")
df = df.set_index("delivery_timestamp")
df.rename(columns={"fixing_i_price_first": "fixing_i_price", "business_date_firs

# choose a specific time window
start, end = "2024-01-01", "2024-03-31"
df_slice = df.loc[start:end]
```

In [ ]:
```
import seaborn as sns
```

```
corr = df_slice.loc[:, (df_slice.columns != "delivery_timestamp") &
                       (df_slice.columns != "business_date")].corr()

plt.figure(figsize=(10, 8))
sns.heatmap(corr, cmap="coolwarm")
plt.title("Correlation Matrix ({} to {})".format(start, end))
plt.show()
```

```
d:\job_interview\hynfra_data_eng_task\.venv\Lib\site-packages\seaborn\matrix.py:5
60: UserWarning: Clustering large matrix with scipy. Installing `fastcluster` may
give better performance.
  warnings.warn(msg)
d:\job_interview\hynfra_data_eng_task\.venv\Lib\site-packages\seaborn\matrix.py:5
60: UserWarning: Clustering large matrix with scipy. Installing `fastcluster` may
give better performance.
  warnings.warn(msg)
<Figure size 1000x800 with 0 Axes>
```

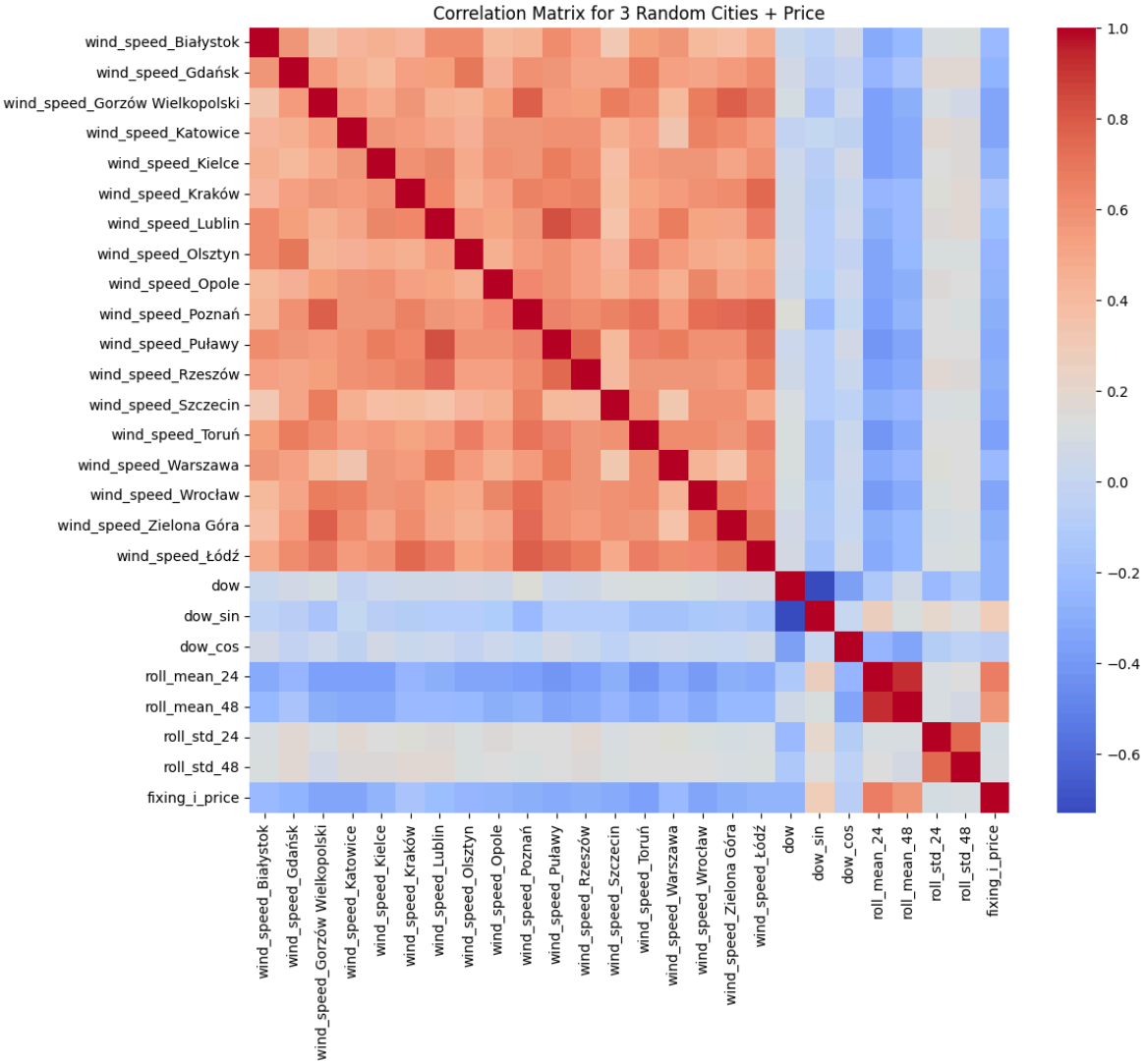


Let's pick 3 random cities to see if there is some sort of correlation with the weather and the price.

In [86]:
```
import numpy as np

# Extract city groups from feature names, excluding the target and date columns
city_feature_cols = [c for c in df_slice.columns if c not in ["fixing_i_price",
city_groups = {}
for col in city_feature_cols:
    parts = col.split("_")
    city = parts[0] if len(parts) > 1 else col
```

```
    city_groups.setdefault(city, []).append(col)

city_names = sorted(city_groups)
random_cities = np.random.default_rng(42).choice(city_names, size=min(3, len(cit
selected_cols = []
for city in random_cities:
    selected_cols.extend(city_groups[city])

# Keep the price target visible in the correlation matrix
if "fixing_i_price" in df_slice.columns:
    selected_cols.append("fixing_i_price")

print("Random cities selected:", list(random_cities))
print("Selected columns for correlation:", selected_cols)

corr_random = df_slice[selected_cols].corr()
plt.figure(figsize=(12, 10))
sns.heatmap(corr_random, cmap="coolwarm")
plt.title("Correlation Matrix for 3 Random Cities + Price")
plt.show()
```

```
Random cities selected: [np.str_('wind'), np.str_('dow'), np.str_('roll')]
Selected columns for correlation: ['wind_speed_Białystok', 'wind_speed_Gdańsk',
'wind_speed_Gorzów Wielkopolski', 'wind_speed_Katowice', 'wind_speed_Kielce', 'wi
nd_speed_Kraków', 'wind_speed_Lublin', 'wind_speed_Olsztyn', 'wind_speed_Opole',
'wind_speed_Poznań', 'wind_speed_Puławy', 'wind_speed_Rzeszów', 'wind_speed_Szcze
cin', 'wind_speed_Toruń', 'wind_speed_Warszawa', 'wind_speed_Wrocław', 'wind_spee
d_Zielona Góra', 'wind_speed_Łódź', 'dow', 'dow_sin', 'dow_cos', 'roll_mean_24',
'roll_mean_48', 'roll_std_24', 'roll_std_48', 'fixing_i_price']
```


Correlation Matrix for 3 Random Cities + Price
wind_speed_Białystok
wind_speed_Gdańsk
wind_speed_Gorzów Wielkopolski
wind_speed_Katowice
wind_speed_Kielce
wind_speed_Kraków
wind_speed_Lublin
wind_speed_Olsztyn
wind_speed_Opole
wind_speed_Poznań
wind_speed_Puławy
wind_speed_Rzeszów
wind_speed_Szczecin
wind_speed_Toruń
wind_speed_Warszawa
wind_speed_Wrocław
wind_speed_Zielona Góra
wind_speed_Łódź
dow
dow_sin
dow_cos
roll_mean_24
roll_mean_48
roll_std_24
roll_std_48
fixing_i_price
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6